Braidyn Sheffield

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EDUCATION

University of Denver, Denver, CO **Expected: June 2026**

Major / Minor: BS in Computer Engineering / Mathematics
GPA: 3.85 / 4.00
Accomplishments: Dean's List (Spring 2023 - Spring 2025)

SKILLS

Technical Skills: Microcontrollers (Arduino, Teensy, STM32), FPGA (Quartus), PCB Design, Circuit Analysis, Soldering, Serial Communication (UART, SPI, I2C), Git/GitHub, 3D Printing, LabVIEW, SOLIDWORKS, Autodesk.
Programming Languages: C, C++ (Arduino, Embedded C), Python, Verilog, MATLAB

PROJECTS

Autonomous Delivery and Identification Vehicle – Team of Five ***March 2025 – June 2025***

- Led a team of 5 to develop an autonomous robotic vehicle for object delivery and color-based identification, aiming to navigate a predefined course without human input.
- Built a modular C++ state machine architecture, improving debugging efficiency by 40% and enabling precise coordination of motors, sensors, and servos.
- Integrated 4 color sensors, 3 ultrasonic sensors, and an MPU-6050 gyro to achieve accurate obstacle detection and 90°/180° turns with < 3° error.
- Engineered a 12V battery-powered system with dual buck converters, ensuring stable 7.2V and 5V outputs and preventing sensor brownouts.
- Successfully achieved fully autonomous navigation, meeting all competition requirements and demonstrating sensor fusion and real-time embedded control.

EKG Circuit Design Project – Team of Two ***May 2024 - June 2024***

- Designed a low-cost analog EKG circuit to capture heart electrical signals as part of a 2-person team addressing real-time biomedical sensing challenges.
- Researched and selected amplifier, filter, and electrode configurations, improving signal clarity and reducing electrical noise by 35%.
- Built and tested the circuit on a breadboard, comparing output waveforms (P, QRS, T waves) to medical-grade references with <5% signal distortion.
- Documented system design, testing, and medical applications in a technical report, earning top evaluation marks from faculty.

PROFESSIONAL EXPERIENCE

Entrepreneurship Garage, Denver CO – *Managing Technical Assistant* ***January 2023 - Present***

- Supervise and train 5 Technical Assistants, ensuring safe equipment use, task delegation, and smooth operation across 3 makerspaces.
- Coordinate and lead 10+ workshops per quarter (Arduino circuits, CAD, 3D printing), directly training 200+ students.
- Assisted students in designing and prototyping 40+ hardware projects using CAD, 3D printing, and microcontroller systems.

US Navy, Atsugi Japan – *Aviation Electronics Technician* ***July 2017 – May 2021***

- Troubleshot and repaired radar, navigation, and communication systems on military aircraft, completing 3000+ maintenance actions and enabling 550+ mishap-free flight hours.
- Led and trained junior technicians in avionics diagnostics, safety procedures, and maintenance documentation.
- Performed pre-flight/post-flight inspections and system calibrations, ensuring aircraft remained fully mission-ready.
- Streamlined tool control and troubleshooting documentation, reducing system downtime and improving inspection readiness.